

4 Mechanics and dynamics

Contents

4.1	Background	49
4.2	Wind Turbine Motions and Loads	50
4.3	General Principles of Mechanics	53
4.4	Methods for modeling Wind Turbine Structural Response	63
4.5	Equations of Motion of Wind Turbine - Discrete and Continuous Systems	64
4.6	Wind Turbine Models Using the Assumed Shape Function Approach	67
4.7	*Linearized Hinge Spring Blade Model	70
4.8	Linearization and Stability	70
4.9	Kinematics	71

Further reading Chapter 4



4.1 Background

Question 4.1: What are the main structural components of a wind turbine

► The typical, and probably simplest, structural components that are studied are:

- Blades
- Shaft
- Tower
- Support structure

The secondary components, are actually more difficult to model and study, and are often sources of failures:

- Pitching system and Blade/hub interface
- Hub
- Shaft bearing
- Gearbox
- Nacelle bed plate
- Yawing system Tower/Nacelle interface
- Tower/Substructure interface
- Foundation

The main components are illustrated in Figure 32.

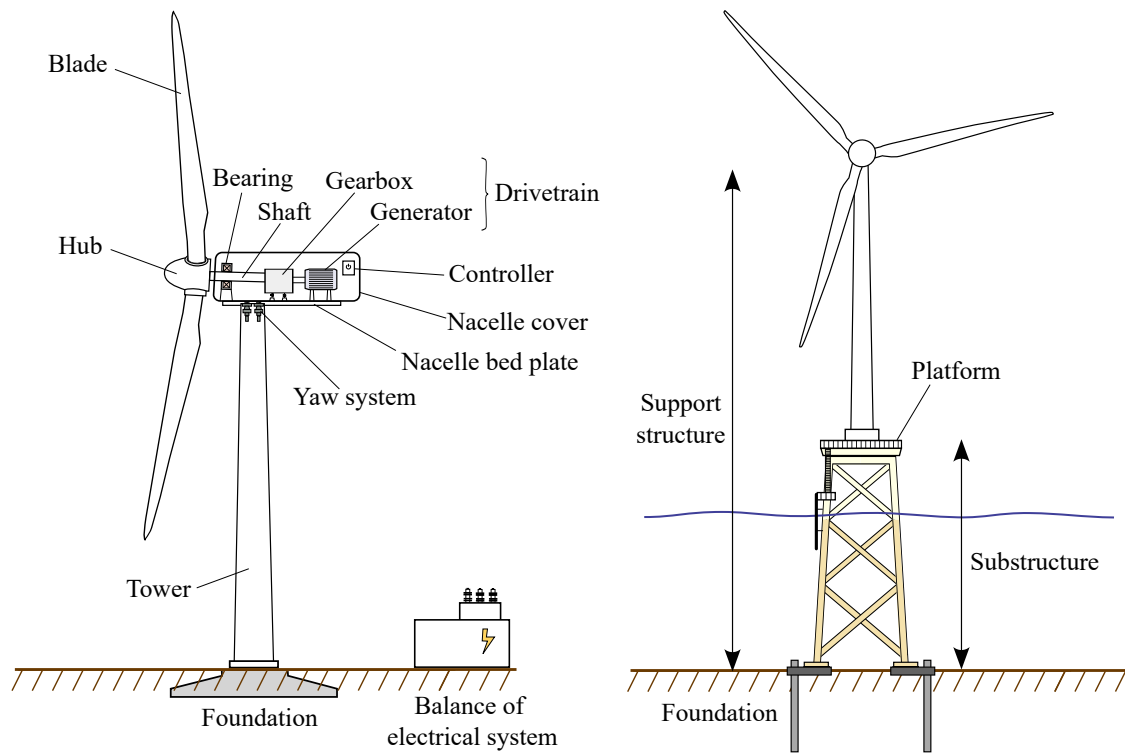


Figure 32: Main wind turbine components

4.2 Wind Turbine Motions and Loads

The typical motions of a wind turbine and degrees of freedom are illustrated in Figure 33.

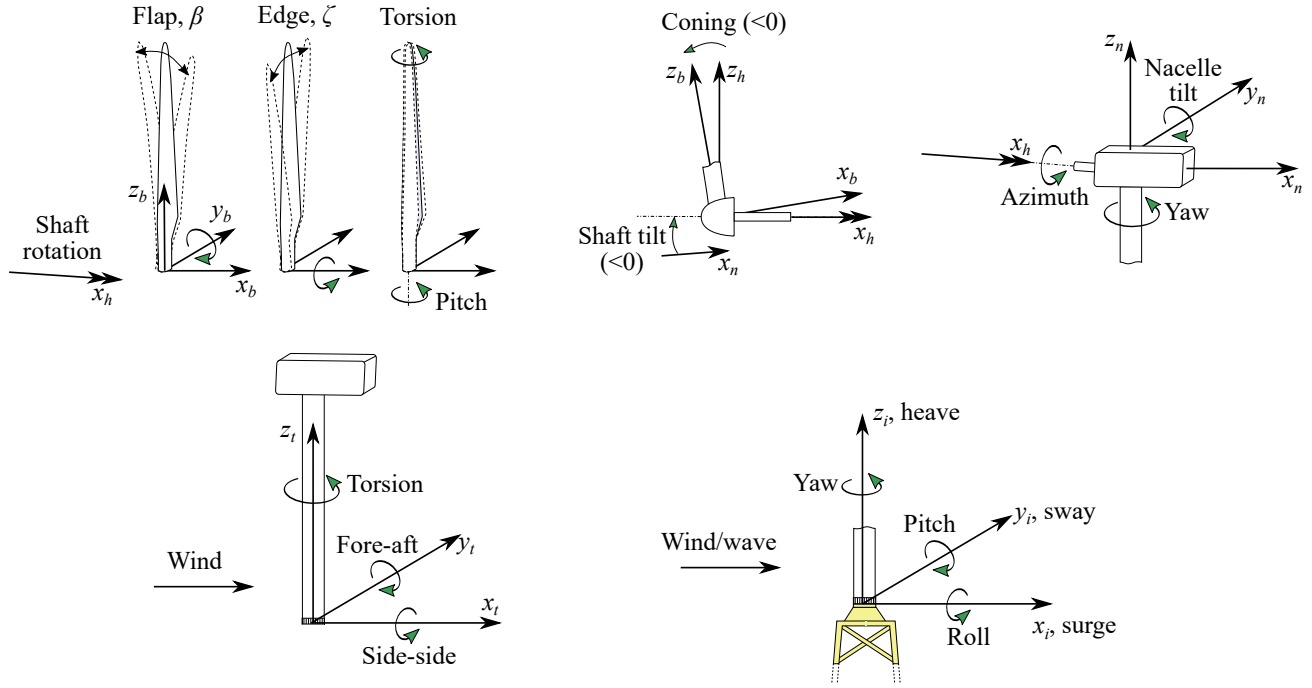


Figure 33: Wind turbine motions / degrees of freedom

The loads acting on a wind turbine are illustrated in Figure 34.

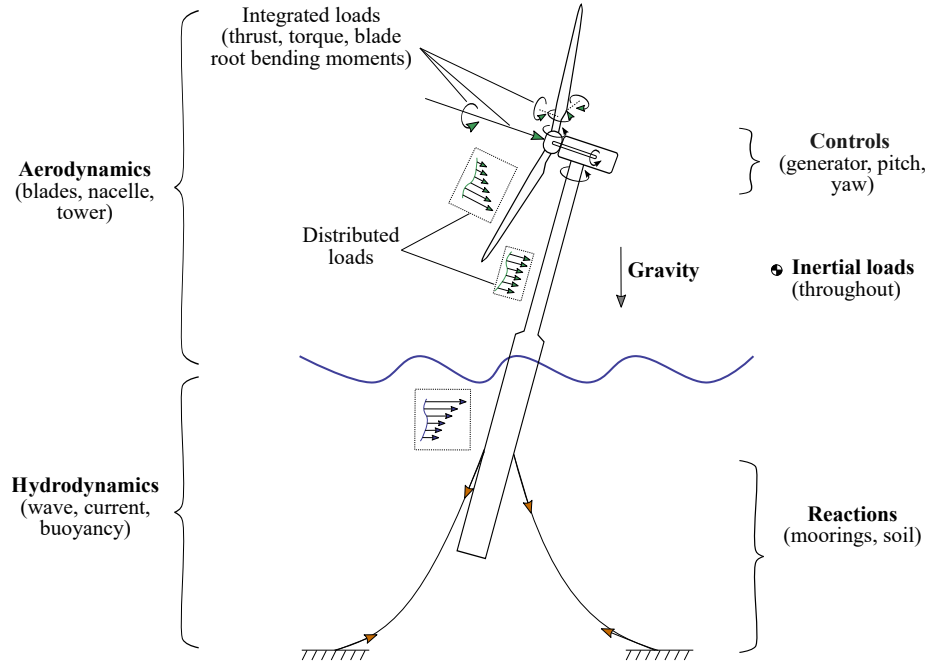


Figure 34: Wind turbine loads

4.2.6 Loads Acting on a Wind Turbine

We write $f_i(s)$ the force per unit length (in [N/m]) on blade i . This force include aerodynamic forces, gravitational, and inertial loads. The curvilinear coordinate s goes along the span of the blade, equal to $s = 0$ at the blade root, and $s = L$ at the blade tip, where L is the integrated curvilinear length, $L = \int_0^L ds$. The notations are illustrated in Figure 35.

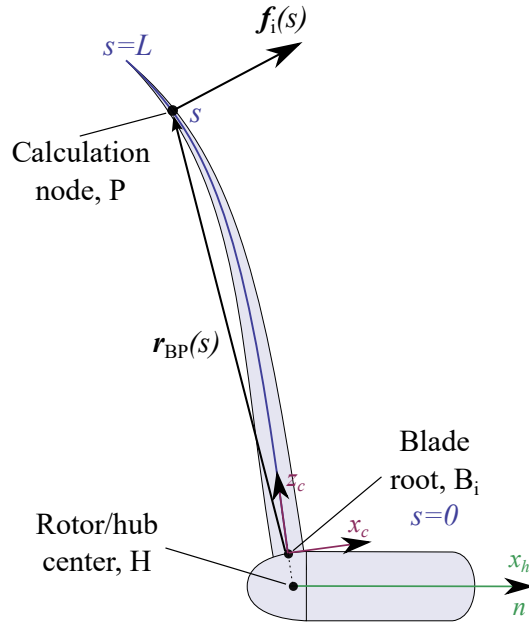


Figure 35: Loads On Blade

For a blade with prebend, presweep, or undergoing elastic deformations, the total force and total torque at the rotor origin (point H) and at the root of blade i (point B_i) are given by:

$$\mathbf{F} = \sum_{i=1}^B \int_0^L \mathbf{f}_i(s) \, ds, \quad \mathbf{M}_H = \sum_{i=1}^B \int_0^L \mathbf{r}_{HP}(s) \times \mathbf{f}_i(s) \, ds, \quad \mathbf{M}_{\text{root},i} \equiv \mathbf{M}_{B_i} = \int_0^L \mathbf{r}_{B_iP}(s) \times \mathbf{f}_i(s) \, ds \quad (4.1)$$

The vectors above may be expressed in any coordinate system. The thrust and torque are obtained by projection against the x axis of the hub ($\hat{\mathbf{x}}_h \equiv \hat{\mathbf{n}}$):

$$T = \mathbf{F} \cdot \hat{\mathbf{x}}_h, \quad Q = \mathbf{M}_H \cdot \hat{\mathbf{x}}_h \quad (4.2)$$

The flapwise and edgewise blade root bending moment are obtained by projection of the blade root bending moment into the “coned” blade root coordinate system: D

$$\mathbf{M}_{\text{flap},i} \triangleq \mathbf{M}_{\text{root},i} \cdot \hat{\mathbf{y}}_c, \quad \mathbf{M}_{\text{edge},i} \triangleq \mathbf{M}_{\text{root},i} \cdot \hat{\mathbf{x}}_c \quad (4.3)$$

The coned coordinate system differs from the blade root coordinate system, noted with a subscript “b”. The blade root system is obtained from the coned system after a rotation corresponding to the blade pitch angle, θ_p , about the z_c axis.

Straight rotor For a straight rotor, using the $r - n - t$ system introduced in Section 3, the blade expands from $r = r_{\text{hub}}$ (the root of the blade) to $r = R$, and the loads are: A

$$T = \sum_{i=1}^B \int_{r_{\text{hub}}}^R f_{n,i}(r) \, dr, \quad Q = \sum_{i=1}^B \int_{r_{\text{hub}}}^R r f_{t,i}(r) \, dr \quad (4.4)$$

The blade root flapwise and edgewise bending moments are:

$$M_{\text{flap},i} = \int_{r_{\text{hub}}}^R (r - r_{\text{hub}}) f_{n,i}(r) \, dr, \quad M_{\text{edge},i} = \int_{r_{\text{hub}}}^R (r - r_{\text{hub}}) f_{t,i}(r) \, dr \quad (4.5)$$

Straight rotor with same loading on each blade For a straight rotor, if the loading is the same on each blade, then from the definition of the local aerodynamic thrust coefficient we get:

$$C_t(r) \triangleq \frac{dT/dr}{\frac{1}{2}\rho U_0^2(2\pi r)} = \frac{B f_n}{\frac{1}{2}\rho U_0^2(2\pi r)} \rightarrow f_n(r) = \frac{1}{2B}\rho U_0^2(2\pi r)C_t(r) \quad (4.6)$$

Therefore:

$$T = B \int_{r_{\text{hub}}}^R f_n(r) dr = \frac{1}{2}\rho U_0^2 \int_{r_{\text{hub}}}^R C_t(r) dr, \quad Q = B \int_{r_{\text{hub}}}^R r f_t(r) dr \quad (4.7)$$

and the flapwise moment is:

$$M_{\text{flap}} = \int_{r_{\text{hub}}}^R (r - r_{\text{hub}}) f_n(r) dr = \frac{\pi}{B}\rho U_0^2 \int_{r_{\text{hub}}}^R (r - r_{\text{hub}}) r C_t(r) dr \quad (4.8)$$

If we add the assumption that the loading is constant along the blade span (i.e., $C_t(r) = C_T$), and there is no hub radius (i.e. $r_{\text{hub}} = 0$), then:

$$M_{\text{flap}} = \frac{2T}{3B}R, \quad M_{\text{edge}} = \frac{Q}{B} \quad (4.9)$$

4.3 General Principles of Mechanics

4.3.1 Selected topics from Mechanics

Newton's laws:

1. First Law (Inertia): A body remains at rest or in uniform motion unless acted upon by an external force.
2. Second Law (Momentum change): The rate of change of momentum of a body is equal to the net force acting on it:

$$\mathbf{f} = \frac{d(m\mathbf{v})}{dt} \quad (4.10)$$

The acceleration of a body is proportional to the net force acting on it:

$$\mathbf{f} = m\mathbf{a} \quad (4.11)$$

3. Third Law (Action & Reaction): For every action, there is an equal and opposite reaction.

4.3.1.1 Coordinate Transformation and Rotations Two coordinate systems can be related to each other using a coordinate transformation matrix of dimension 3×3 . This is typically relevant when going from a coordinate system attached to a body (noted b) to the global coordinate system (noted g). It is recalled that a vector is an entity that exists independently of the coordinate system, but its components are different when expressed in different coordinate systems. The vector $\mathbf{v}$ is written $\mathbf{v}_b = (v_x, v_y, v_z)$ in the body coordinate system and $\mathbf{v}_g = (v_x, v_y, v_z)$ in the global coordinate system. The two systems are related with the 3×3 transformation matrix $\mathbf{R}_{gb}$ using:

$$\mathbf{v}_g = \mathbf{R}_{gb}\mathbf{v}_b \quad (4.12)$$

A 3×3 matrix $\mathbf{M}$ is transformed from the body to a global system as:

$$\mathbf{M}_g = \mathbf{R}_{gb} \mathbf{M}_b \mathbf{R}_{gb}^T \quad (4.13)$$

where T denotes the transpose operator. In this text, we use bold typeface for matrices and vectors, and uppercase letters for matrices whenever possible.

The coordinate transformation can be constructed using successive rotations, and the transformation matrix can be obtained as a product of elementary rotation matrices. Examples of representations using three successive rotations are the Euler angles or Bryant angles representations. The use of three angles can become singular, and general-purpose code needs four parameters to represent transformation matrices, such as the Euler parameters. In this book, only simple transformations are used. If the body frame is rotated by an angle θ around the x axis compared to the global system, the coordinate transformation is given by:

$$\mathbf{R}_{gb} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \quad (4.14)$$

The matrix given above is the elementary rotation matrix about the x axis. Similar matrices are obtained about the other axes.

4.3.1.3 Area moment of inertia For an arbitrary surface S and an arbitrary axis D , the second moment of area is defined as: D

$$I_\Delta = \int_S d^2(\mathbf{x}, \Delta) \, dS \quad [m^4] \quad (4.15)$$

where $d(\mathbf{x}, \Delta)$ is the distance between the point $\mathbf{x}$ and the axis Δ , and $d\mathbf{x}$ is an elementary surface (typically $dx \, dy$). The section S is contained in a plane in most engineering applications. The planar and polar second moment of area defined below are definitions commonly used as special case of Equation 4.15.

The parallel axis theorem can be used to transfer the area moment of inertia from the axis Δ to another axis, Δ' , parallel to it, as:

$$I_{\Delta'} = I_\Delta + A d^2(\Delta, \Delta') \quad (4.16)$$

The term planar second moment of area is used when the axis Δ is coplanar with the plane containing the surface S . For a section S defined in the $x - y$ plane, the planar second moment of area with respect to the x and y axis are: D

$$S \in x\text{-}y \text{ plane,} \quad I_x = \iint_S y^2 \, dx \, dy, \quad I_y = \iint_S x^2 \, dx \, dy \quad (4.17)$$

For a section S defined in the $y - z$ plane, the planar second moment of area with respect to the y and z axis are:

$$S \in y\text{-}z \text{ plane,} \quad I_y = \iint_S z^2 \, dy \, dz, \quad I_z = \iint_S y^2 \, dy \, dz \quad (4.18)$$

The term polar second moment of area is used when the axis Δ is perpendicular to the plane containing the surface S . The term *polar moment of inertia* is commonly used, but it not recommended since it introduces confusion with the mass moment of inertia. There is a link between the two when the mass density of a cross section is constant, see below. D

The polar moment is sometimes written J or I_p . The latter notation will be used. For a section S defined in the $x - y$ plane, the polar second moment of area is defined about the axis z , while for a section in the $y - z$ plane,

it is defined about the axis x , that is:

$$S \in x\text{-}y \text{ plane, } I_p = I_z = \iint_S (x^2 + y^2) dx dy \quad [\text{m}^4] \quad (4.19)$$

$$S \in y\text{-}z \text{ plane, } I_p = I_x = \iint_S (y^2 + z^2) dy dz \quad [\text{m}^4] \quad (4.20)$$

It is readily seen that the polar second moment of area can be obtained as the sum of the two planar moment of area.

Examples 4.1: For an annulus in the $x - y$ plane of outer radius r_2 and inner radius r_1 (see Figure 36), the planar second moment of area is:

$$S \in x\text{-}y \text{ plane, } I_x = I_y = \frac{1}{4}[r_2^4 - r_1^4], \quad I_p = I_z = \frac{\pi}{2}[r_2^4 - r_1^4] = 2I_x \quad (4.21)$$

In terms of diameter D and inner diameter $D - t$, the planar second moment of area is:

$$I_x = I_y = \frac{\pi}{64}[D^4 - (D - t)^4] \approx \frac{\pi}{8}D^3t, \quad I_p = I_z = \frac{\pi}{32}[D^4 - (D - t)^4] = 2I_x \quad (4.22)$$

where the approximation is obtained using a first order Taylor expansion ($t \ll D$).

For a rectangular $x - y$ -cross section of height h and length b (see Figure 36), the planar second moment of area is:

$$S \in x\text{-}y \text{ plane, } I_x = \frac{bh^3}{12}, \quad I_y = \frac{b^3h}{12}, \quad I_p = I_z = \frac{bh}{12}(b^2 + h^2) \quad (4.23)$$

If the area moment of inertia are expressed at the point C instead of the point O , then:

$$S \in x\text{-}y \text{ plane, } I_{x,C} = \frac{bh^3}{3}, \quad I_{y,C} = \frac{b^3h}{3} \quad (4.24)$$

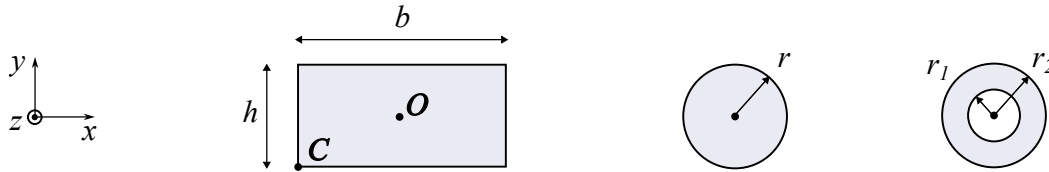


Figure 36: Example of cross sections for moment of inertia calculation.

In practice, we can compute the area moment of inertia by discretizing a cross section, and computing the integral as a sum over small area, accounting for the distance to the axis. This is illustrated in Figure 37.

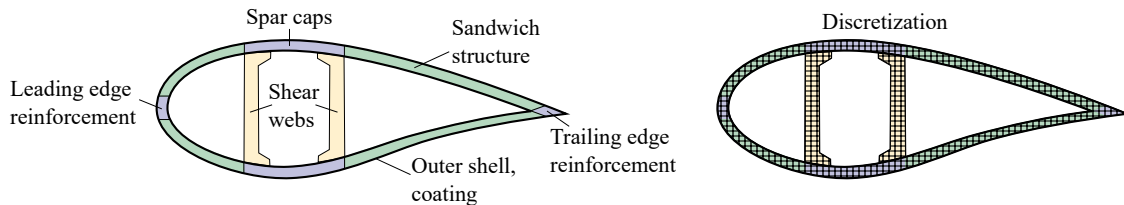


Figure 37: Airfoil cross section made of different material properties (of different density and strength), and its discretization for numerical integration of the inertial and stiffness properties

4.3.1.4 Mass Moment of Inertia The moment of inertia of a volume $\mathcal{V}$ about an axis Δ centered at a given point P :

D

$$J_{\Delta} \triangleq \int_{\mathcal{V}} d^2(\mathbf{x}, \Delta) dm(\mathbf{x}) = \int_{\mathcal{V}} d^2(\mathbf{x}, \Delta) \rho(\mathbf{x}) d\mathcal{V} \quad [\text{kg} \times \text{m}^2] \quad (4.25)$$

where $d\mathbf{x} = d\mathcal{V}$ is an elementary volume (typically $dx dy dz$). If the axis is along $\mathbf{e}_x$, and centered in the origin, then $d(\mathbf{x}, \Delta) = \sqrt{y^2 + z^2} = r$, and hence:

$$J_{xx,O} = \int_{\mathcal{V}} (y^2 + z^2) \rho(x, y, z) dx dy dz \quad (4.26)$$

The parallel axis theorem can be used to transfer the mass moment of inertia from the axis Δ to another axis, Δ' , parallel to it, as:

$$J_{\Delta'} = J_{\Delta} + M d^2(\Delta, \Delta') \quad (4.27)$$

Examples 4.2: For a uniform cylinder of length l and radius r , along the z axis, of mass $m = \rho\pi r^2 l$, with origin O , and center of mass G :

$$J_{xx,G} = J_{yy,G} = \frac{ml^2}{12}, \quad J_{xx,O} = J_{yy,O} = J_{yy,G} + m(l/2)^2 = \frac{ml^2}{3}, \quad (4.28)$$

$$J_{zz,O} = J_{zz,G} = \frac{\pi\rho l r^4}{2} = m \frac{r^2}{2} \quad (4.29)$$

The term planar moment of inertia is used to refer to the mass moment of inertia of a cross section of a body. Similar to the mass moment of inertia, it is defined about an axis Δ centered at a given point P and given by:

D

$$J_{p,\Delta} = \int_S [\text{distance}(\mathbf{x}, \Delta)]^2 dm(\mathbf{x}) = \int_S [\text{distance}(\mathbf{x}, \Delta)]^2 \rho(\mathbf{x}) d\mathbf{x} \quad [\text{kg} \times \text{m}] \quad (4.30)$$

For a cross section of constant density ρ , the planar moment of inertia is directly proportional to the polar moment of area:

$$J_{p,\Delta} = \rho I_{p,\Delta} \quad [\text{kg} \times \text{m}] \quad (4.31)$$

Examples 4.3: For an annulus in the $x-y$ plane of outer radius R and inner radius $R-t$ and of constant density ρ (mass $m = \rho\pi(R^2 - (R-t)^2)$), the planar moment of inertia is:

$$J_{xx,O} = J_{yy,O} = \frac{1}{4} m [R^2 - (R-t)^2] \quad (4.32)$$

$$J_{zz,O} = \frac{1}{2} m [R^2 - (R-t)^2] \quad (4.33)$$

Skew symmetric matrix The skew symmetric matrix of a vector $\mathbf{u}$ is defined as:

D

$$[\tilde{\mathbf{u}}] \triangleq \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}, \quad [\tilde{\mathbf{u}}] = -[\tilde{\mathbf{u}}]^T \quad (4.34)$$

It can be used to express the cross product of two vectors:

$$[\tilde{\mathbf{u}}] \mathbf{v} \equiv \mathbf{u} \times \mathbf{v} \quad (4.35)$$

The skew symmetric matrix is often used in structural dynamics. For instance, the quantity $-\mathbf{[\tilde{u}]}^2$ appears in the definition of the inertia tensor (see below). It can readily shown that:

$$-\mathbf{[\tilde{u}]}^2 \equiv -\mathbf{[\tilde{u}]} \mathbf{[\tilde{u}]} = \mathbf{[\tilde{u}]}^T \mathbf{[\tilde{u}]} = \|\mathbf{u}\|^2 \mathbf{I} - \mathbf{u} \mathbf{u}^T = \begin{bmatrix} u_y^2 + u_z^2 & -u_x u_y & -u_x u_z \\ -u_y u_x & u_x^2 + u_z^2 & -u_y u_z \\ -u_z u_x & -u_z u_y & u_x^2 + u_y^2 \end{bmatrix} \quad (4.36)$$

4.3.1.5 Inertia tensor To express the laws of mechanics of a rigid body, it is necessary to obtain its inertia tensor about its center of gravity, $\mathbf{J}_G$, as it is used to express the angular momentum $\mathbf{H}_G$ and the kinetic energy $\mathcal{T}$ of a body as:

$$\mathbf{H}_G = \mathbf{J}_G \boldsymbol{\omega}, \quad \mathcal{T} = \frac{1}{2} \boldsymbol{\omega}^T \mathbf{J}_G \boldsymbol{\omega} \quad (4.37)$$

The inertia tensor of a body defined by a volume $\mathcal{V}$, expressed at the origin O , is given by:

D

$$\mathbf{J}_O = \begin{bmatrix} J_{xx} & J_{xy} & J_{xz} \\ J_{yx} & J_{yy} & J_{yz} \\ J_{zx} & J_{zy} & J_{zz} \end{bmatrix} \triangleq - \int_{\mathcal{V}} [\tilde{\mathbf{r}}_{OP}]^2 dm = \int_{\mathcal{V}} \rho \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -yx & x^2 + z^2 & -yz \\ -zx & -zy & x^2 + y^2 \end{bmatrix} d\mathcal{V} \quad (4.38)$$

The off-diagonal terms are referred to as products of inertia. They are obtained as:

D

$$J_{xy} = J_{yx} = - \int_{\mathcal{V}} \rho xy d\mathcal{V}, \quad J_{xz} = J_{zx} = - \int_{\mathcal{V}} \rho xz d\mathcal{V}, \quad J_{yz} = J_{zy} = - \int_{\mathcal{V}} \rho yz d\mathcal{V} \quad (4.39)$$

Properties:

- **Change of frame** The values of the inertia matrix depends on the frame in which it is expressed. The inertia matrix values are transformed using basis transformation matrices. Typically, the inertia matrix with respect to point O is expressed in the body frame (b), $\mathbf{J}|_b$, and it is converted to an inertial frame (i) using the transformation matrix $\mathbf{R}_{ib}$ from body coordinates to (i)-coordinates as follows:

$$\mathbf{J}_O|_i = \mathbf{R}_{ib} \mathbf{J}_O|_b \mathbf{R}_{ib}^T, \quad \mathbf{J}_O|_b = \mathbf{R}_{ib}^T \mathbf{J}_O|_i \mathbf{R}_{ib} \quad (4.40)$$

- **Principal axes** The inertia tensor is real, symmetric and positive definite. It has three distinct eigenvectors that can be used to define a basis for which the tensor is in a diagonal form, referred to as the basis of the principle axes, noted (p). In this basis, the inertial tensor is diagonal:

$$\mathbf{R}_{ip} \mathbf{J}_O|_i \mathbf{R}_{ip}^T = \begin{bmatrix} J_{\xi} & 0 & 0 \\ 0 & J_{\eta} & 0 \\ 0 & 0 & J_{\zeta} \end{bmatrix} = \mathbf{J}_O|_p \quad (4.41)$$

where $\mathbf{R}_{ip}$ is the transformation matrix from the principle axes to the reference frame. Rotating the body around one of the principle axes will result in an angular momentum (angular momentum $\mathbf{H} = \mathbf{J}\boldsymbol{\omega}$) along this axis only.

- **Change of reference point** The inertia matrix is a function of the reference point. The change of reference point can be simplified using the value of the inertia matrix about the center of mass. The vector $\mathbf{r}_{OP}$ is expressed using the center of gravity as $\mathbf{r}_{OP} = \mathbf{r}_{OG} + \mathbf{r}_{GP}$, leading to: $[\tilde{\mathbf{r}}_{OP}]^2 = [\tilde{\mathbf{r}}_{OG}]^2 + 2[\tilde{\mathbf{r}}_{OG}] [\tilde{\mathbf{r}}_{GP}] + [\tilde{\mathbf{r}}_{GP}]^2$. Inserting this decomposition into Equation 4.38 leads to:

$$\begin{aligned} \mathbf{J}_O &\triangleq - \int_{\mathcal{V}} [\tilde{\mathbf{r}}_{OP}]^2 dm = -M [\tilde{\mathbf{r}}_{OG}]^2 + 0 - \int_{\mathcal{V}} [\tilde{\mathbf{r}}_{GP}]^2 dm \\ \mathbf{J}_O &= \mathbf{J}_G - M [\tilde{\mathbf{r}}_{OG}]^2 \end{aligned} \quad (4.42)$$

Given two points A and B , and G the center of mass of the body:

$$\mathbf{J}_B = \mathbf{J}_A - M \left([\tilde{\mathbf{r}}_{BG}]^2 - [\tilde{\mathbf{r}}_{AG}]^2 \right) \quad (4.43)$$

If the point A is the center of gravity, $\mathbf{r}_{AG} = \mathbf{0}$ and Equation 4.42 is obtained.

The symmetry of three

Example 4.4: The inertia tensor of a blade can be computed from the definitions above, for instance, choosing x as the rotor axis, and with y and z forming an arbitrary orthonormal basis with the x axis. The inertia tensors of the other blades in this coordinate system are obtained by multiplying the matrix of blade 1 on both sides by rotation matrices of angles $2\pi/3$ and $4\pi/3$, respectively, about the x axis. The inertia of the rotor is then obtained as:

$$\mathbf{J}_{\text{rotor},O} = \begin{bmatrix} 3J_{xx} & 0 & 0 \\ 0 & \frac{3}{2}(J_{yy} + J_{zz}) & 0 \\ 0 & 0 & \frac{3}{2}(J_{yy} + J_{zz}) \end{bmatrix} \quad (4.44)$$

It is observed that the moments of inertia are the same for the y and z axis, which were chosen arbitrarily. Therefore, the inertia tensor has the same matricial representation for any choice of $y - z$ axes: it remains a diagonal matrix, with the same diagonal quantities, irrespective of the azimuthal position of the rotor. The inertia tensor of a three-bladed rotor with identical blades has the same property as the one of a uniform disk: the (mass) moment of inertia about any axis in the plane normal to the rotor axis is the same. 

The result applies to a rotor with more than three blades, replacing 3 in Equation 4.44 by the number of blades. The result does not hold for a rotor with 1 or 2 blades, for which the inertia tensor will be periodic with the azimuth and will be a function of the choice of the y and z axes. 

4.3.2 Bending of a Cantilever beam

Further reading See Section G

Please read Section G. 

4.3.3 Rigid Body (Planar Rotation)

Further reading See textbook

The book talks about planar rotation, for which, Newton's law writes:

$$J_P \ddot{\theta} = \tau_P \quad (4.45)$$

If z is the axis normal to the plane, then $J_P = J_{zz,P}$. Note: all points of a rigid body have the same rotational velocity and acceleration (but not the same linear velocities and acceleration).

Planar dynamics For a rigid body, moving in a plane, the unconstrained equation of motions are:

$$\begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & J_P \end{bmatrix}_i \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{\theta} \end{bmatrix}_i = \begin{bmatrix} f_x \\ f_y \\ \tau_P \end{bmatrix}_i \quad (4.46)$$

Rigid body dynamics - Newton-Euler equations Linear motion and rotational motion of a body is given by the Newton-Euler equation. In the inertial frame, and at the center of mass, they are given by:

$$m\ddot{\mathbf{r}}^G = \mathbf{F} \quad (4.47)$$

$$\frac{d\mathbf{H}^G}{dt} = [\mathbf{J}^G]\dot{\omega} + [\tilde{\omega}][\mathbf{J}^G]\omega = \boldsymbol{\tau}^G, \quad \mathbf{H}^G = [\mathbf{J}^G]\omega \quad (4.48)$$

They are written in matricial form as follows:

$$\text{(At COG, inertial frame)} \quad \begin{bmatrix} m[\mathbf{I}] & 0 \\ 0 & [\mathbf{J}^G] \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{r}}^G \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \boldsymbol{\tau}^G - [\tilde{\omega}][\mathbf{J}]\omega \end{bmatrix} \quad (4.49)$$

If the reference point is not at the center of mass $\boldsymbol{\rho} = \mathbf{r}_{AG}$:

$$\text{(All at A, inertial frame)} \quad \begin{bmatrix} m[\mathbf{I}] & -m[\tilde{\rho}] \\ m[\tilde{\rho}] & [\mathbf{J}^A] \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{r}}^A \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} \mathbf{F} - m[\tilde{\omega}][\tilde{\omega}]\boldsymbol{\rho} \\ \boldsymbol{\tau}^A - [\tilde{\omega}][\mathbf{J}^A]\omega \end{bmatrix} \quad (4.50)$$

Question 4.2: What is the term: $[\tilde{\omega}][\mathbf{J}]\omega$? When is it zero?

► This term represents the Gyroscopic forces. It comes from the derivation of the angular momentum $\mathbf{H} = \mathbf{J}\omega$. This term is $\omega \times \mathbf{H}$. It represents the forces that occur on a spinning body when the angular momentum and the rotational speed are not parallel. For planar cases, it is always zero. In 3D cases, it might be zero if the rotational speed is about one of the principal axes of the body, in which case the angular momentum remains parallel to ω .

4.3.4 Gears and gear ratio

Further reading See textbook

We consider two shafts separated by a gearbox. Both sides of the gearbox have their own rotational speed Ω , moment of inertia J and torque Q . The notations are illustrated in Figure 38. We write $n = n_{\text{gear}}$ the gear ratio relating the speeds of the drive train at A and B, and η is the efficiency of the gearbox. Quantities from one side are related to the other side as follows:

$$\Omega_B = \Omega_A \cdot n \quad (4.51)$$

$$Q_A = Q_B \cdot n \cdot \eta \quad (4.52)$$

$$\Omega_A Q_A = \Omega_B Q_B \cdot \eta \quad (4.53)$$

Quantities can be transferred to the other side as follows:

$$J_{B,\text{side A}} = J_B \cdot n^2 \quad (4.54)$$

$$J_{A,\text{side B}} = J_A / n^2 \quad (4.55)$$

$$J_{\text{tot,side A}} = J_A + J_B \cdot n^2 \quad (4.56)$$

$$J_{\text{tot,side B}} = J_B + J_A / n^2 \quad (4.57)$$

If the shafts are not rigid, it is also possible to convert their stiffness and combine them (see textbook):

$$k_{B,\text{side A}} = k_B \cdot n \quad (4.58)$$

$$k_{A,\text{side B}} = k_A / n \quad (4.59)$$

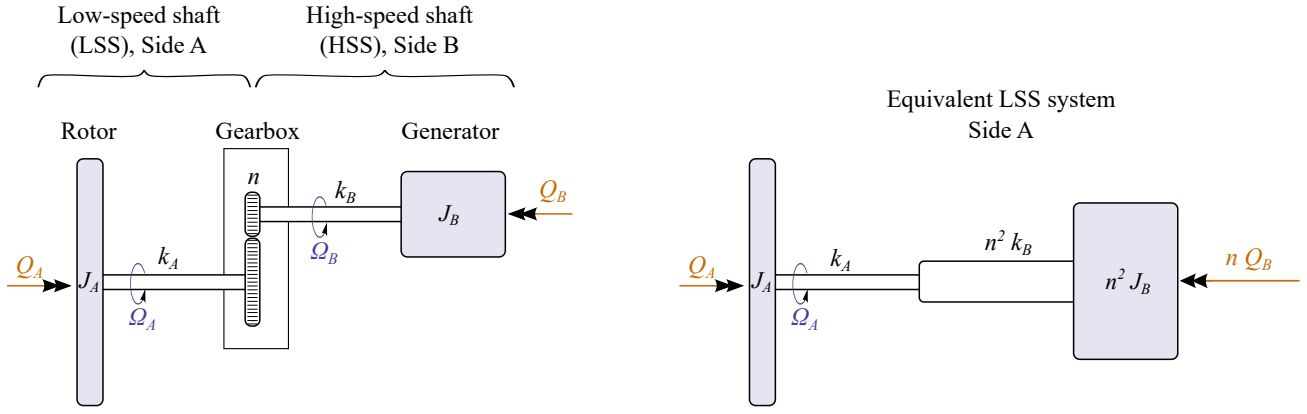


Figure 38: Geared system representing a wind turbine rotor and generator separated by a gearbox; J, inertia; K, stiffness; n, speed ratio between the shafts, and system equivalent to geared system.

4.3.5 *Inertia loads from rotations (Gyroscopic Motion)

Further reading See textbook

Note: the book focuses on gyroscopic.

When rotations are involved, the term “ ma ” is not as straightforward. It requires us to compute the inertia tensor, and the rotational motion have additional contributions to the linear acceleration. The acceleration contributions are a function of the generalized velocities $\dot{x}$, therefore these contributions are sometimes referred to as quadratic velocities.

We recall the following derivation of the acceleration from one coordinate system (or body) to another:

$$\mathbf{r}_P = \mathbf{r}_0 + \mathbf{r}' \quad (4.60)$$

$$\mathbf{v}_P = \mathbf{v}_0 + \boldsymbol{\omega} \times \mathbf{r}' + \dot{\mathbf{r}}' \quad (4.61)$$

$$\mathbf{a}_P = \mathbf{a}_0 + \underbrace{\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}')}_{\text{Centrifugal}} + \dot{\boldsymbol{\omega}} \times \mathbf{r}' + \underbrace{2\boldsymbol{\omega} \times \dot{\mathbf{r}}'}_{\text{Coriolis}} + \ddot{\mathbf{r}}' \quad (4.62)$$

The contributions are:

- Centrifugal loads due to centripetal acceleration. Given a body rotating at $\boldsymbol{\omega}$, and $\mathbf{r}'$ a point of the body expressed with respect to its origin, the centrifugal load is:

$$-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}') \quad (4.63)$$

- Coriolis load. If a point moves with a velocity $\mathbf{v}_{\text{rel}}$ relative to a rotating coordinate system, it experiences the Coriolis load:

$$-2\boldsymbol{\omega} \times \mathbf{v}_{\text{rel}} \quad (4.64)$$

- Gyroscopic load. If body 2 is connected to body 1, the rotational axis of body 2 will be rotated by body 1, leading to the following gyroscopic load on body 1:

$$-m\dot{\boldsymbol{\omega}} \times \mathbf{r}', \quad \dot{\boldsymbol{\omega}} = \boldsymbol{\omega}_1 \times \boldsymbol{\omega}_2 \quad (4.65)$$

Question 4.3: When does gyroscopic loads matter for wind turbines and why?

► Gyroscopic effects are always present for wind turbines, but they are strongest when we try to yaw the turbine (we want to change the direction of the angular momentum, and it is resisting us).

4.3.6 Vibrations: Single DOF

Further reading See textbook

The canonical example in mechanical engineering is the mass-spring-damper system, excited by an input force. The system is illustrated in ??.

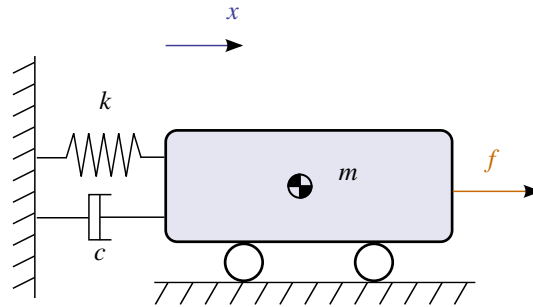


Figure 39: Mass spring damper system

The equation of motion is:

$$m\ddot{x} + c\dot{x} + kx = f(t) \quad (4.66)$$

m , c , k are constants, representing the mass, damping and stiffness, and $f(t)$ is the external force. This is a second-order linear-time-invariant (LTI) system (the highest-derivative is of second-order, it is linear, and the coefficients are constant). The equation of motion is:

$$m\ddot{x} + c\dot{x} + kx = f \quad (4.67)$$

$$\ddot{x} + 2\frac{c}{2m}\dot{x} + \frac{k}{m}x = \frac{f}{m} \quad (4.68)$$

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2x = \frac{f}{m} \quad (4.69)$$

The last form is the “canonical” form which introduces the natural frequency, ω_0 , and the damping ratio, ζ . Various relations can be obtained between the physical variables (m , c , k) and these variables:

D

$$\omega_0 = \sqrt{\frac{k}{m}}, \quad \zeta \triangleq \frac{c}{2m\omega_0} \equiv \frac{c\omega_0}{2k} \equiv \frac{c}{2\sqrt{km}} = \frac{c}{c_c} \quad (4.70)$$

where c_c , is the critical damping, defined as:

D

$$c_c \triangleq 2\sqrt{km} = 2m\omega_0 = \frac{2k}{\omega_0} \quad (4.71)$$

The damped frequency is the actual frequency at which the system will oscillate, it is defined as:

D

$$\omega_d \triangleq \omega_0 \sqrt{1 - \zeta^2} \quad T_d \triangleq \frac{1}{f_d} = \frac{2\pi}{\omega_d} \quad (4.72)$$

where T_d the damped period. Using “matrix notations”:

$$qm q = 1, \quad \Leftarrow \quad q = \sqrt{\frac{1}{m}}, \quad \lambda = \frac{k}{m}, \quad \beta_{ii} = qcq = q^2c = \frac{c}{m} \quad (4.73)$$

The logarithmic decrement is defined as the ratio between the amplitude of two oscillations:

D

$$\delta \triangleq \frac{1}{N} \log \frac{x(t)}{x(t + NT_d)} \quad (4.74)$$

For a second order system, the following can be shown:

$$\delta = \frac{2\pi\zeta}{\sqrt{1 - \zeta^2}}, \quad \zeta = \frac{1}{\sqrt{1 + \left(\frac{2\pi}{\delta}\right)^2}} = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \approx \frac{\delta}{2\pi} \quad (4.75)$$

Laplace Transform The Laplace transform of ?? is readily obtained as:

$$s^2 X(s) - sx_0 - \dot{x}_0 + 2\zeta\omega_0 s X + \omega_0^2 X(s) = bU(s) \quad (4.76)$$

$$X(s) = \underbrace{\frac{bU(s)}{s^2 + 2\zeta\omega_0 s + \omega_0^2}}_{\text{forced}} + \underbrace{\frac{(s + 2\zeta\omega_0)x_0 + \dot{x}_0}{s^2 + 2\zeta\omega_0 s + \omega_0^2}}_{\text{free}} \quad (4.77)$$

Setting the initial conditions to zero, the transfer function from state the inputs is:

$$G(s) \triangleq \frac{X(s)}{U(s)} = \frac{b}{s^2 + 2\zeta\omega_0 s + \omega_0^2} \quad (4.78)$$

Looking at Equation 4.77, it is apparent that the time domain solution will be the sum of a free-response (which depends on the initial condition but not on the input u), and a forced-response (which depends on u but not on the initial conditions). ⚠

Based on Equation 4.78, the poles of the system (roots of the denominator) are obtained using the quadratic formula as:

$$p_i = -\zeta\omega_0 \pm \omega_0 \sqrt{\zeta^2 - 1} = -\zeta\omega_0 \pm j\omega_0 \sqrt{1 - \zeta^2} \quad (4.79)$$

We immediately see that if ζ is positive, then $\text{Re}(p_i) < 0$ and the system is stable.

For a stable system, the natural frequency and damping ratio are obtained from the pole as:

$$\omega_0 = |p|, \quad \zeta = -\frac{\text{Re}(p)}{\omega_0} \quad (4.80)$$

Example: $x(t) = Ae^{-\alpha t} \cos(\omega_0 t)$, i.e $x(t + nT_d) = x(t)e^{-\alpha nT_d}$:

$$\delta = \frac{1}{n} \log(e^{\alpha nT_d}) = \alpha T_d, \quad \alpha = \frac{\delta}{T_d} = \frac{\delta}{2\pi} \omega_d = \zeta\omega_0 \quad (4.81)$$

4.3.7 Rotational Equations of Motion and Vibration

Further reading See textbook

The textbook provide the example illustrated in Figure 40, which can be used to model the torsion of the shaft if a break is applied on the rotor (for instance after an emergency shutdown). The equation of motion is:

$$J_r \ddot{\psi} + k_{DT} \psi = Q_a \quad (4.82)$$

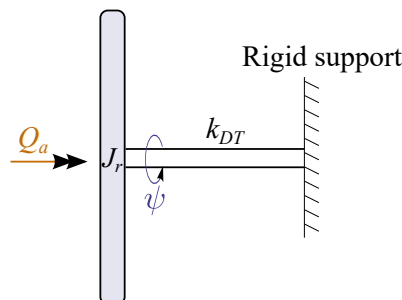


Figure 40: Model of a rotor with shaft torsion.

For a rigid rotor, the natural frequencies of the drive train torsion in free-free (generator on) and fixed-free (shaft cannot rotate or rotates at constant speed) conditions are:

$$\omega_{\text{free-free}} = \sqrt{\frac{k_{\text{DT}}}{J_{\text{rot}}} + \frac{k_{\text{DT}}}{J_g n_g^2}} \quad (4.83)$$

$$\omega_{\text{fixed-free}} = \sqrt{\frac{k_{\text{DT}}}{J_{\text{rot}}}} \quad (4.84)$$

With flexible blades the frequencies are likely to change. If the edgewise motions move in-phase with the rotational motion, the drivetrain natural frequency is likely to decrease relative to the rigid-rotor case. But if the edgewise motions move out-of-phase with the rotational motion, the drivetrain natural frequency is likely to increase relative to the rigid-rotor case.

Exercise 4.1: Find the equation of motion for the drive train represented in Figure 38. ■

4.3.8 Introduction to Vibration of Cantilevered Beams

Further reading See textbook and Section G

Please read Section G.



4.4 Methods for modeling Wind Turbine Structural Response

4.4.1 Lumped Parameter Method

You can skip it.

4.4.2 Modal Analysis Method and Shape Function Approach

Further reading See textbook



4.4.3 Finite Element Method

If you are curious.

4.4.4 Multibody Analysis

You can skip it.

4.4.5 Multiphysics Model

You can skip it.

4.5 Equations of Motion of Wind Turbine - Discrete and Continuous Systems

4.5.1 Derivation of the Equations of Motion - Lagrange Method (Not in Book)

Introduction to Lagrangian Mechanics Lagrangian mechanics provides a powerful and elegant method to analyze mechanical systems. By following a systematic procedure, one can derive equations of motion for even complex systems efficiently.

Generalized coordinates A system's configuration is uniquely described by a minimal set of n generalized coordinates, denoted as $\{q_1, q_2, \dots, q_n\}$.

Lagrange's Equations The Euler-Lagrange equation is derived from the principle of least action, leading to:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_{i,\text{nc}}, \quad \mathcal{L} = \mathcal{T} - \mathcal{V} \quad (4.85)$$

where $\mathcal{T}$ is the kinetic energy, $\mathcal{V}$ is the potential energy of the system and where $Q_{i,\text{nc}}$ are generalized forces that arise from dissipative effects, external inputs, or reaction forces due to constraints.

Generalized forces The generalized forces can be obtained by analyzing the virtual work $\delta W_i = Q_i \delta q_i$ for each virtual displacement δq_i .

Introducing dissipative energy Non-conservative forces that are dissipative are sometimes gathered into a dissipation energy $\mathcal{D}$, such that the dissipative forces are obtained as $Q_{i,\text{dis}} = \partial \mathcal{D} / \partial \dot{q}_i$. Introducing the definition of $\mathcal{T}$, $\mathcal{V}$ and $\mathcal{D}$ into Equation 4.85 leads to:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{T}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{T}}{\partial q_i} + \frac{\partial \mathcal{V}}{\partial q_i} + \frac{\partial \mathcal{D}}{\partial \dot{q}_i} = Q_i \quad (4.86)$$

Expressions for Kinetic, Potential, and Dissipative Energy The kinetic energy is given by the following forms:

$$\mathcal{T} = \frac{1}{2}mv^2, \quad \mathcal{T} = \frac{1}{2}J\omega^2, \quad \mathcal{T} = \frac{1}{2}\omega^T \mathbf{J}_G \omega, \quad \mathcal{T} = \frac{1}{2} \int v(\mathbf{x})^2 dm(\mathbf{x}), \quad \mathcal{T} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} \quad (4.87)$$

Conservative forces derive from the potential energy $\mathcal{V}$ ($\mathbf{f} = -\nabla \mathcal{V}$). Typical example include gravity or elastic forces. Examples are given below:

$$\mathcal{V} = mgz, \quad \mathcal{V} = \int gz(\mathbf{x}) dm(\mathbf{x}), \quad \mathcal{V} = \frac{1}{2}kx^2, \quad \mathcal{V} = \int EI(z) \left[\frac{d^2 u}{dz^2} \right]^2 dz, \quad \mathcal{V} = \frac{1}{2} \mathbf{q}^T \mathbf{K} \mathbf{q} \quad (4.88)$$

Typical examples of dissipative energy are linear dampers, examples of which are given below:

$$\mathcal{D} = \frac{1}{2}cv^2, \quad \mathcal{D} = \frac{1}{2}c_\theta \omega^2, \quad \mathcal{D} = \frac{1}{2} \int c(\mathbf{x})v(\mathbf{x})^2 dm(\mathbf{x}), \quad \mathcal{D} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{C} \dot{\mathbf{q}} \quad (4.89)$$

Procedure to Derive Equations of Motion The procedure to apply Lagrange's method is as follows:

1. Define the system's degrees of freedom and choose appropriate generalized coordinates q_i .
2. Express the kinetic energy $\mathcal{T}$ and potential energy $\mathcal{V}$ in terms of q_i and $\dot{q}_i$.
3. Construct the Lagrangian $\mathcal{L} = \mathcal{T} - \mathcal{V}$.
4. Compute the derivatives $\frac{\partial \mathcal{L}}{\partial q_i}$ and $\frac{\partial \mathcal{L}}{\partial \dot{q}_i}$.
5. Apply the Euler-Lagrange equation to obtain the equations of motion.
6. Solve the resulting equations to describe the system's dynamics.

Example - Simple Pendulum A pendulum of length l and mass m has a single generalized coordinate: the angle θ . The kinetic and potential energies are:

$$\mathcal{T} = \frac{1}{2}m(l\dot{\theta})^2 = \frac{1}{2}ml^2\dot{\theta}^2, \quad (4.90)$$

$$\mathcal{V} = -mgl \cos \theta. \quad (4.91)$$

The Lagrangian is:

$$\mathcal{L} = \frac{1}{2}ml^2\dot{\theta}^2 + mgl \cos \theta. \quad (4.92)$$

Applying the Euler-Lagrange equation gives:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0 \quad (4.93)$$

$$ml^2\ddot{\theta} + mgl \sin \theta = 0 \quad (4.94)$$

Example of Lagrange Equation applied to wind turbine models

Further reading See Section H

Also, see Section M (HW4).



4.5.2 Equations of Motion for Multiple Degrees of Freedom System

Further reading See textbook



4.5.3 Examples of Rigid Body Wind Turbine Models with Few Degrees of Freedom

Further reading See textbook and Section H and Section M. Also checkout [3].



Figure 41

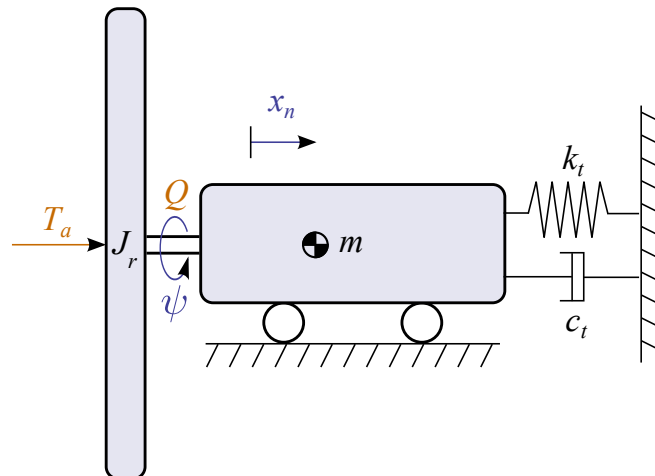


Figure 41: Model of a wind turbine, using tower-top displacement (x_n) and shaft rotation (ψ) as degrees of freedom - See textbook.

Figure 42

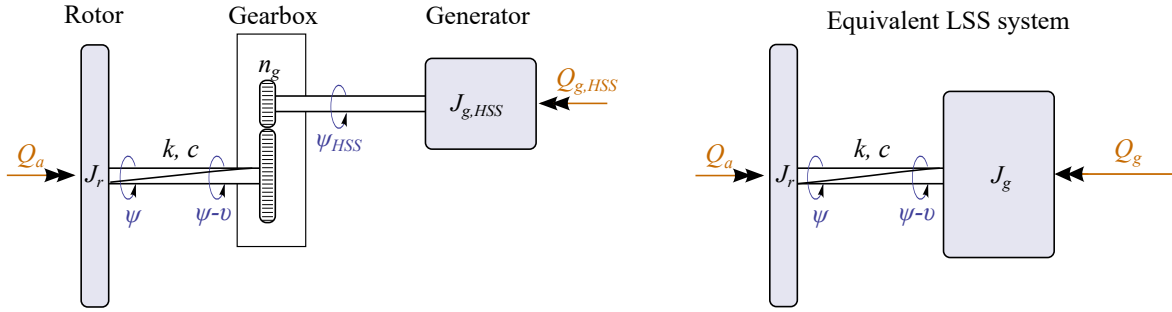


Figure 42: Model of a wind turbine drivetrain with shaft torsion - See textbook.

4.5.4 Solution of the System of Equations

Further reading See textbook

Make sure you know how to write a system into its first-order state-space form.

4.5.5 Continuous Systems

Further reading

- See Section G which illustrate how to compute deflections and mode shapes for a beam
- See application of Lagrange equation to a continuous system in Section H
- Note: the textbook present the Micklestad method, you can study it if you want, it should return the same results as the method presented in Section G.
- More applications follow in the next section Section 4.6

Figure 43

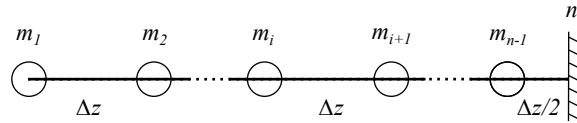


Figure 43: Lumped mass model of cantilevered beam; m_i , mass; Δz , distance between masses; n , number of stations

Figure 44

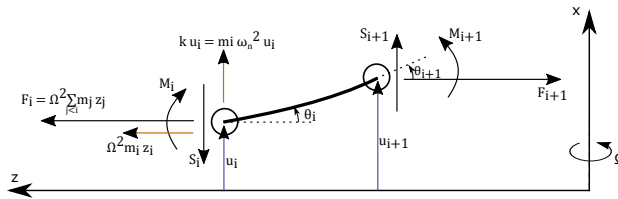


Figure 44: Free body diagram of a beam section for a rotating beam using the Micklestad method.

4.6 Wind Turbine Models Using the Assumed Shape Function Approach

4.6.1 EOM for an Isolated Beam Described Using One Shape Function

Further reading See Textbook

68

In the shape function approach, we start by assuming a given form for the displacement field. Here, we assume that the field $u(z, t)$ in the x direction is given by a single shape function. The notations are illustrated in Figure 45.

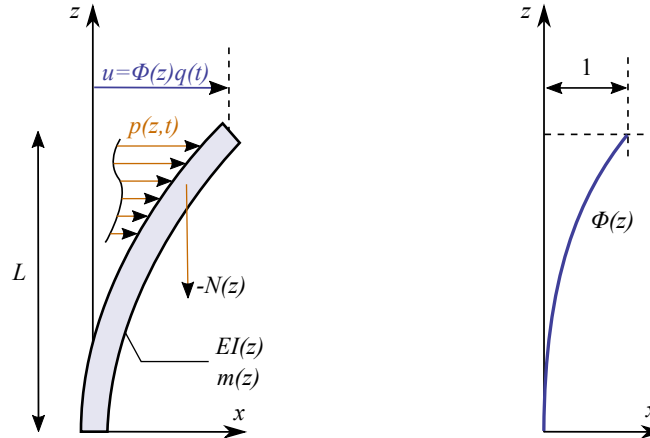


Figure 45: Sketch of a clamped-free beam. Its displacement is represented using one shape function

The displacement field is given by:

$$u(z, t) = \Phi(z)q(t) \quad (4.95)$$

$$\ddot{u}(z, t) = \Phi(z)\ddot{q}(t) \quad (4.96)$$

where $\Phi(z)$ is the shape function and $q(t)$ is the generalized coordinate.

EOM using Lagrange

$$\mathcal{T} = \int_0^L m(z) \|\mathbf{v}\|^2 dz \quad \mathbf{v}(z) = \frac{d\mathbf{r}(z)}{dt}, \quad \mathbf{r} = \mathbf{r}_O + \mathbf{r}_{OP_0} + \mathbf{u} \quad (4.97)$$

$$\mathcal{V} = \int_0^L EI(z) \left[\frac{d^2 u}{dz^2} \right]^2 dz \quad (4.98)$$

EOM using variational / weak formulation The equations of motions are obtained by inserting Equation 4.95 into Equation G.7, multiplying by the shape function, $\Phi(z)$, and then integrating, giving:

$$M\ddot{q}(t) + (K_{EI} + K_N)q(t) = Q(t) \quad (4.99)$$

$$M = \int_0^L m(z)\Phi(z)^2 dz \quad (4.100)$$

$$K_{EI} = \int_0^L EI(z) \left[\frac{d^2 \Phi(z)}{dz^2} \right]^2 dz \quad (4.101)$$

$$K_N = \int_0^L N(z) \left[\frac{d\Phi(z)}{dz} \right]^2 dz \quad (4.102)$$

$$N(z) = \int_z^L p_z(z) dz \quad (4.103)$$

$$Q(t) = \int_0^L \Phi(z)p(z, t) dz \quad (4.104)$$

Various applications follow below (and in the textbook).

The natural frequency is obtained as:

$$\omega_0 = \sqrt{\frac{K}{M}} \quad (4.105)$$

Generalized damping may be introduced based on the damping ratio (see Section 4.3.6):

$$C = 2M\omega_0\zeta \quad (4.106)$$

4.6.2 Illustration of the Tower Softening Effect Using a Beam and a Top Mass

Further reading See Textbook

68

Example 4.5: The impact of the axial load on the tower natural frequency is studied in this example. The notations are illustrated in Figure 46.

The influence of the top mass on the generalized stiffness is obtained using $N_t(z) = -M_t g$:

$$K_{N,t} = -M_t g \int_0^L \left[\frac{d\Phi}{dz} \right]^2 dz \quad (4.107)$$

The influence of the self-weight is obtained using $p_z(z) = -mg$:

$$N_w(z) = - \int_z^L m(z') g dz' \quad (4.108)$$

$$K_{N,w} = \int_0^L N_w(z) \left[\frac{d\Phi}{dz} \right]^2 dz = \int_0^L \int_z^L m(z') g dz' \left[\frac{d\Phi}{dz} \right]^2 dz \quad (4.109)$$

In the textbook, an assumed mode shape of $\Phi(z) = 1 - \cos(\frac{\pi z}{2L})$ is used so that closed-form solutions can be obtained.

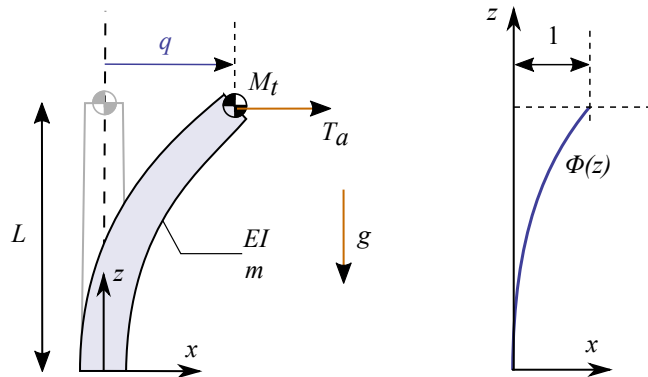


Figure 46: One-DOF model of a tower with top mass, using the Rayleigh-Ritz approach with one shape function

4.6.3 Illustration of the Centrifugal Stiffening

Further reading See Textbook

68

Example 4.6: The notations are illustrated in Figure 47.

The axial loading from the centrifugal force acting on a blade is given by:

$$N(r) = \int_r^R r' \Omega^2 m(r') dr', \quad p_r(r) = r \Omega^2 m(r) \quad (4.110)$$

The corresponding loading in the transverse direction of the beam as:

$$p_N(r) = N(r) \frac{d^2 u}{dr^2} - m r \Omega^2 \frac{du}{dr} \quad (4.111)$$

The corresponding generalized stiffness is

$$K_N(\Omega) = \int_0^R N(r) \left[\frac{d\Phi}{dr} \right]^2 dr = \Omega^2 \int_0^R \int_r^R m(r') r' dr' \left[\frac{d\Phi}{dr} \right]^2 dr \quad (4.112)$$

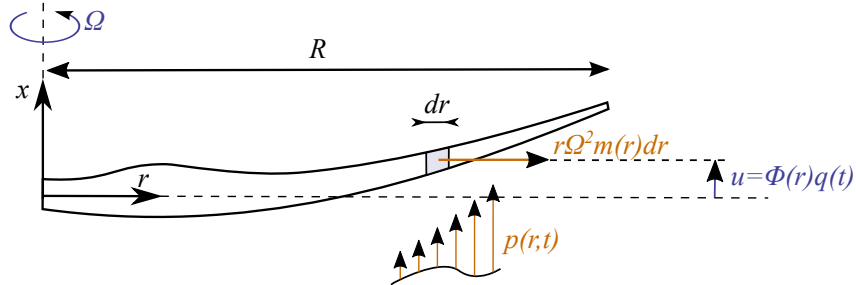


Figure 47: Centrifugal stiffening

4.6.4 *The Assumed Shape Function Approach with Multiple Shape Functions

Further reading

- See textbook, slightly more advanced, but you have the basis if you understand it for one shape function.
- See [3] for more background and examples of Wind Energy applications.



$$\mathbf{u}(z, t) = \sum_{i=1}^N \Phi_i(z) q_i(t) \quad (4.113)$$

$$\dot{\mathbf{u}}(z, t) = \sum_{i=1}^N \Phi_i(z) \dot{q}_i(t) \quad (4.114)$$

$$\ddot{\mathbf{u}}(z, t) = \sum_{i=1}^N \Phi_i(z) \ddot{q}_i(t) \quad (4.115)$$

The kinetic energy is:

$$\mathcal{T} = \frac{1}{2} \int_0^L m \dot{u}^2 dz = \frac{1}{2} \sum_i \sum_j M_{ij} \dot{q}_i \dot{q}_j \quad (4.116)$$

where M_{ij} is:

$$M_{ij} = \int_0^L m(z) \Phi_i(z) \cdot \Phi_j(z) dz, \quad i, j = 1, \dots, n \quad (4.117)$$

Similarly, the potential energy (strain energy) of the beam is obtained as

$$\mathcal{V} = \frac{1}{2} \sum_i \sum_j K_{ij} q_i q_j \quad (4.118)$$

where K_{ij} are the element of the stiffness matrix, which, under the assumption of small deformations, are given by:

$$K_{ij} = \int_0^L [EI_x(z) \kappa_{i,y}(z) \kappa_{j,y}(z) + EI_y \kappa_{i,x} \kappa_{j,x}] dz, \quad i, j = 1, \dots, n \quad (4.119)$$

where $\kappa_{i,y} = \frac{d^2 \Phi_{i,x}}{dz^2}$ is the y-curvature of the shape function i . Elongation and torsional strains (EA and GK_t) can similarly be added to the strain energy.

The external loads on the beam are assumed to consist of a distributed force vector $\mathbf{p}(z)$. The virtual work done by the force $\mathbf{p}$ for each virtual displacement δq_i , provides the generalized force as: $Q_i = \int_0^L \Phi_i \cdot \mathbf{p} dz$. The equations of motion are obtained using Lagrange's equation: $\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} = Q_i$, for $i = 1..n$, and written in matricial form as:

$$\mathbf{M} \ddot{\mathbf{q}} + \mathbf{K} \mathbf{q} = \mathbf{Q} \quad (4.120)$$

where $\mathbf{q} = [q_1, \dots, q_n]$ and the elements of the generalized mass and stiffness matrices are given in Equation 4.117 and Equation 4.119. Damping is typically added a posteriori to the equations, where the Rayleigh damping assumption is often used $\mathbf{D} = \alpha \mathbf{M} + \beta \mathbf{K}$. If the shape functions are mode shapes, then the mass and stiffness matrices are diagonal, and the stiffness values would be $K_{ii} = \omega_i^2 M_{ii}$, with ω_i the eigenfrequency of the beam mode i .

If the beam is also loaded axially by a force $N(z)$, then from Equation G.7 it appears that this force produces a distributed load in the transverse direction equal to $\mathbf{n} = \frac{\partial}{\partial z} [N(z) \frac{\partial \mathbf{u}}{\partial z}]$, with components in the y and z direction. The generalized force associated with this loading is then: $Q_{i,N} = \int_0^L \Phi_i \cdot \mathbf{n} dz$. Inserting the expression of $\mathbf{n}$ and $\mathbf{u}$, the generalized force has the form of a stiffness term: $Q_{i,N} = - \sum_j K_{ij,N} q_j$ with

$$K_{ij,N} = - \int_0^L \Phi_i \cdot \frac{d}{dz} \left[N(z) \frac{d\Phi_j}{dz} \right] dz = \int_0^L N(z) \frac{d\Phi_i}{dz} \cdot \frac{d\Phi_j}{dz} dz - \left[N(x) \Phi_i \cdot \frac{d\Phi_j}{dz} \right]_0^L \quad (4.121)$$

and where integration by parts was used to obtain the second equality.

4.7 *Linearized Hinge Spring Blade Model

You can skip it.

4.8 Linearization and Stability

Further reading See Textbook



Metrics When doing prescribed motion, aeroelastic stability can be assessed by computing the work done by the aerodynamic force over one cycle. A stable system will have positive work done.

$$W = \oint_{\text{cycle}} \mathbf{F} \cdot d\mathbf{x} \quad (4.122)$$

For a pure plunging motion, the prescribed motion would be $x(t) = A \sin \omega t$.

4.9 Kinematics

4.9.1 Kinematics

In kinematics, the evolution of the DOFs in time are assumed prescribed, and we determine the motion (position/speed/acceleration) of given points or of the whole structure. D

4.9.2 Notations

For a given body b , and a point P belonging to the body the notations introduced in Table 5 are introduced.

Table 5: Different kinematic notations

Explicit	Nikravesh	Description
$\mathbf{v}$	NA	Vector (could be expressed in any frame)
$\mathbf{v} _i$	$\mathbf{v}$	Vector expressed in the inertial coordinate system
$\mathbf{v} _b$	$\mathbf{v}'$	Vector expressed in the body coordinate system
$\mathbf{r}_{OB}$	$\mathbf{r}_b$	Vector from the global origin to body origin (in the inertial coordinate system)
$\mathbf{r}_{OP}$	$\mathbf{r}_P$	Vector from the global origin to point P of a body (in the inertial coordinate system)
$\mathbf{r}_{BP}$	$\mathbf{s}_P$	Vector from the body origin B to point P (in the inertial coordinate system)
$\mathbf{r}_{BP} _b$	$\mathbf{s}'_P$	Vector from the body origin B to point P in the body coordinate system
$\mathbf{r}_{BP_0} _i$	$\mathbf{s}_{P_0}$	Vector from the body origin B to undeflected point P_0 in the inertial coordinate system
$\mathbf{r}_{P_0P} _i$	$\mathbf{u}_P$	Vector from the undeflected point P_0 to the deflected point P in the inertial coordinate system
$\mathbf{R}_{ib} = \mathbf{R}$	$\mathbf{A}$	Transformation matrix from body to inertial coordinates
$\boldsymbol{\omega}_b = \boldsymbol{\omega}$	$\boldsymbol{\omega}_b = \boldsymbol{\omega}$	Angular velocity vector of the body with respect to the inertial frame.
$(\cdot)_b$	$\cdot'$	Time derivative in the body frame

Derivative in a moving frame Let $\mathbf{v}$ be a time-dependent vector. The time derivative of $\mathbf{v}$ observed in the inertial frame (i) and the body frame (b) are related by

$$\left(\frac{d\mathbf{v}}{dt} \right)_i = \left(\frac{d\mathbf{v}}{dt} \right)_b + \boldsymbol{\omega}_b \times \mathbf{v},$$

where $\boldsymbol{\omega}_b$ is the angular velocity of the body frame with respect to the inertial frame. This relation accounts for the apparent change in $\mathbf{v}$ due to the rotation of the body frame. A consequence of the above, is that the time derivative of the transformation matrix is given by:

$$\dot{\mathbf{A}} = [\tilde{\boldsymbol{\omega}}]\mathbf{A} \quad (4.123)$$

Explicit notations

$$\mathbf{r}_{OP} = \mathbf{r}_{OB} + \mathbf{r}_{BP} = \mathbf{r}_{OB} + \mathbf{r}_{BP_0} + \mathbf{r}_{P_0P} \quad (4.124)$$

$$\mathbf{v}_{OP} = \mathbf{v}_{OB} + \boldsymbol{\omega} \times \mathbf{r}_{BP} + (\dot{\mathbf{r}}_{P_0P})_b \quad (4.125)$$

Typically, the vectors $\mathbf{r}_{BP_0}$ and $\mathbf{r}_{P_0P}$ are more easily expressed in the body coordinates. The velocity is therefore obtained in the inertial frame as:

$$\mathbf{v}_{OP}|_i = \mathbf{v}_{OB}|_i + \boldsymbol{\omega}|_i \times \mathbf{R}_{ib} \mathbf{r}_{BP}|_b + (\dot{\mathbf{r}}_{P_0P})_b \quad (4.126)$$

Nikravesh notations

$$\mathbf{r}_P = \mathbf{r}_b + \mathbf{s}_P = \mathbf{r}_b + \mathbf{s}_{P_0} + \mathbf{u}_P \quad (4.127)$$

$$\mathbf{v}_P = \mathbf{v}_b + \boldsymbol{\omega} \times \mathbf{s}_P + (\dot{\mathbf{u}}_P)_b \quad (4.128)$$

$$\mathbf{a}_P = \mathbf{a}_b + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{s}_P) + \dot{\boldsymbol{\omega}} \times \mathbf{s}_P + 2\boldsymbol{\omega} \times (\dot{\mathbf{u}}_P)_b + (\ddot{\mathbf{u}}_P)_b \quad (4.129)$$

Typically, the vectors $\mathbf{s}$ and $\mathbf{u}$ are more easily expressed in the body coordinates. For instance, the velocity is:

$$\mathbf{v}_P = \dot{\mathbf{r}}_b + \dot{\mathbf{s}}_P = \dot{\mathbf{r}}_b + [\tilde{\boldsymbol{\omega}}]\mathbf{s}_P + \mathbf{A}\dot{\mathbf{s}}'_P = \dot{\mathbf{r}}_b + \dot{\mathbf{s}}_P = \dot{\mathbf{r}}_b + [\tilde{\boldsymbol{\omega}}]\mathbf{s}_P + \mathbf{A}(\dot{\mathbf{u}}'_P)_b \quad (4.130)$$

Relations between bodies Given a body i and a body j , we use the subscript ij for position/rotations between body i and j , and therefore we have:

$$\mathbf{r}_j = \mathbf{r}_i + \mathbf{r}_{ji} = \mathbf{r}_{ji} + \mathbf{r}_i \quad (4.131)$$

$$\mathbf{A}_j = \mathbf{A}_{ji}\mathbf{A}_i \quad (4.132)$$

$$\boldsymbol{\omega}_j = \boldsymbol{\omega}_i + \boldsymbol{\omega}_{ji} \quad (4.133)$$

Rigid body Points A and B belong to the same rigid body, of angular velocity $\boldsymbol{\omega}$, angular acceleration $\dot{\boldsymbol{\omega}}$:

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{AB} \quad (4.134)$$

$$\dot{\mathbf{r}}_B = \dot{\mathbf{r}}_A + \boldsymbol{\omega} \times \mathbf{r}_{AB} \quad (4.135)$$

$$\ddot{\mathbf{r}}_B = \ddot{\mathbf{r}}_A + \dot{\boldsymbol{\omega}} \times \mathbf{r}_{AB} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{AB}) \quad (4.136)$$

Spatial kinematics - Notes

- An unconstrained body in space requires 6 independent quantities to determine its configuration: 3 translational and 3 rotational components
- The choice of rotational coordinates is not unique (Euler angles, Euler parameters, Rodriguez parameters, direction cosines, etc), and algebraic relations relate them when more than 3 coordinates are used. Representation using 3 variables will suffer from singularity at certain orientations of the frame. Indeed, three angles are not enough information since you need to know in which order the rotations are done.
- The transformation matrix (or direction cosine matrix) $\mathbf{A}$ contains 9 components, subject to 6 constraints (coming from $\mathbf{A}\mathbf{A}^t = \mathbf{I}$), and thus only 9-6=3 independent coordinates.
- Euler's theorem: The general displacement of a body with one point fixed is a rotation about some axis, called the *orientational axis of rotation*. Note: this is an imaginary axis of rotation, and it is not the *instantaneous axis of rotation* of the body (along $\boldsymbol{\omega}$).
- Change of orientation can be interpreted in two different ways[?, p. 157] (the algebra is the same, but the rotation is thought with different signs):
 - active point of view: the matrix $\mathbf{A}$ is a *rotation matrix*, an operator that relates two vectors, expressed in the same (global) coordinate system. This is the approach employed by Euler parameters to obtain $\mathbf{A}$.
 - passive point of view: the matrix $\mathbf{A}$ is a *transformation matrix*, an operator that relates the coordinates of one vector in two different coordinate systems. This is the approach employed by Bryant or Euler angles to obtain $\mathbf{A}$.
- The angular rotation of a rigid body is unique to a body (independent of location on the body)

- When performing successive rotations, the order of rotations is essential and cannot be reversed ($\mathbf{A}_1\mathbf{A}_2 \neq \mathbf{A}_2\mathbf{A}_1$), i.e. rotations are non-commutative (unless they are about parallel axes).
- Given the coordinates of 3 points in global coordinates, two on the body axes and one at the origin, one can readily get the transformation matrix $\mathbf{A}$ [?, p. 164].
- Given 3 rotations, θ , or 4 Euler parameters $\mathbf{p}$ (one direction vector and one angle), one can get the transformation matrix $\mathbf{A}$ and $\mathbf{G}$

$$\theta \rightarrow (\mathbf{A}, \mathbf{G}), \quad \omega = \mathbf{G}\dot{\theta} \quad (4.137)$$

- Inversely, given the matrix $\mathbf{A}$, one can obtain the Euler parameters $\mathbf{p}$ (the magnitudes are determined using the trace of $\mathbf{A}$ and its diagonal elements, but the signs are determined using off-diagonal elements) [?, p 161]. The inverse relationship from $\mathbf{A}$ to the Euler parameters is never singular, but it can be singular with other set of angles such as Bryant or Euler angles.

Euler theorem: The general displacement of a body with one point fixed is a rotation about some axis.

Chasles's theorem: The most general displacement of a body is a translation plus a rotation.

- Transformation matrix $[A]$: passive point of view, expressing the coordinates of a vector in two different coordinate systems, $s = [A]s'$ and $s' = [A]^t s$
- Rotation matrix $[A]$: active point of view, rotating a vector while the coordinate system stays the same: $t^* = [A]t$

Irrespectively of the rotational coordinate representation θ (Euler parameters, Euler angles, Bryant angles..), one can obtain the $\mathbf{A}$ matrix, and matrices $\mathbf{G}$ and $\bar{\mathbf{G}}$ such that:

$$\omega = \mathbf{G}\dot{\theta}, \quad \omega' = \bar{\mathbf{G}}\dot{\theta}, \quad \dot{\theta} = \mathbf{G}^{-1}\omega = \bar{\mathbf{G}}^{-1}\omega' \quad (4.138)$$

$$\dot{\omega} = \mathbf{G}\ddot{\theta}, \quad \dot{\omega}' = \bar{\mathbf{G}}\ddot{\theta}, \quad (4.139)$$

For 3-parameters representations, the inverse matrices are singular for a given angle. For the Euler parameter representation, the $\mathbf{G}$ -matrices are 3×4 and the inverse is a pseudo-inverse.

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